

First 10 Rows - Raw Data

Obs	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize	Make
1	A1	2017	12500	Manual	15735	Petrol	150	55.4	1.4	audi
2	A6	2016	16500	Automatic	36203	Diesel	20	64.2	2	audi
3	A1	2016	11000	Manual	29946	Petrol	30	55.4	1.4	audi
4	A4	2017	16800	Automatic	25952	Diesel	145	67.3	2	audi
5	A3	2019	17300	Manual	1998	Petrol	145	49.6	1	audi
6	A1	2016	13900	Automatic	32260	Petrol	30	58.9	1.4	audi
7	A6	2016	13250	Automatic	76788	Diesel	30	61.4	2	audi
8	A4	2016	11750		75185	Diesel	20	70.6	2	audi
9	A3	2015	10200	Manual	46112	Petrol	20	60.1	1.4	audi
10	A1	2016	12000	Manual	22451		30	55.4	1.4	audi

Dataset Contents - Raw

The CONTENTS Procedure

Data Set Name	WORK.CARS_RAW	Observations	72435
Member Type	DATA	Variables	10
Engine	V9	Indexes	0
Created	09/09/2026 17:51:48	Observation Length	72
Last Modified	09/09/2026 17:51:48	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	40
First Data Page	1
Max Obs per Page	1816
Obs in First Data Page	1773
Number of Data Set Repairs	0
Filename	/saswork/SAS_work90A50001DCD9_odaws01-euw1.oda.sas.com/SAS_work97CA0001DCD9_odaws01-euw1.oda.sas.com/cars_raw.sas7bdat
Release Created	9.0401M9
Host Created	Linux
Inode Number	2642415
Access Permission	rw-r--r--
Owner Name	u64509872
File Size	5MB
File Size (bytes)	5373952

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
10	Make	Char	4	\$4.	\$4.
9	engineSize	Num	8	BEST12.	BEST32.
6	fuelType	Char	6	\$6.	\$6.
5	mileage	Num	8	BEST12.	BEST32.
1	model	Char	2	\$2.	\$2.
8	mpg	Num	8	BEST12.	BEST32.
3	price	Num	8	BEST12.	BEST32.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
7	tax	Num	8	BEST12.	BEST32.
4	transmission	Char	9	\$9.	\$9.
2	year	Num	8	BEST12.	BEST32.

Numerical Features - Before Cleaning

The MEANS Procedure

Variable	N	N Miss	Mean	Std Dev	Minimum	Maximum	Median
year	68813	3622	2017.08	2.0994742	1996.00	2020.00	2017.00
price	68814	3621	16583.92	9304.41	495.0000000	145000.00	14495.00
mileage	68813	3622	23179.44	21347.57	1.0000000	323000.00	17521.00
tax	68814	3621	116.9304211	64.0819309	0	580.0000000	145.0000000
mpg	68814	3621	55.8363516	17.0436431	0.3000000	470.8000000	55.4000000
engineSize	68813	3622	1.6353727	0.5617338	0	6.6000000	1.6000000

Missing Values per Column

The MEANS Procedure

Variable	N Miss
year	3622
price	3621
mileage	3622
tax	3621
mpg	3621
engineSize	3622

Categorical Frequencies - Raw Data

The FREQ Procedure

Make	Frequency	Percent	Cumulative Frequency	Cumulative Percent
BMW	10269	14.92	10269	14.92
Ford	17045	24.77	27314	39.69
Hyun	4606	6.69	31920	46.39
audi	10137	14.73	42057	61.12
skod	5947	8.64	48004	69.76
toyo	6417	9.33	54421	79.08
vw	14393	20.92	68814	100.00
Frequency Missing = 3621				

fuelType	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Diesel	27446	39.88	27446	39.88
Electr	5	0.01	27451	39.89
Hybrid	2747	3.99	30198	43.88
Other	225	0.33	30423	44.21
Petrol	38390	55.79	68813	100.00
Frequency Missing = 3622				

transmission	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Automatic	13358	19.41	13358	19.41
Frequency Missing = 3623				

transmission	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Manual	40839	59.35	54197	78.76
Other	4	0.01	54201	78.77
Semi-Auto	14611	21.23	68812	100.00
Frequency Missing = 3623				

Missing Values - After Cleaning

The MEANS Procedure

Variable	N	N Miss
year	43325	0
price	43325	0
mileage	43325	0
tax	43325	0
mpg	43325	0
engineSize	43325	0

Missing Values - After Cleaning

rows_before	rows_after	rows_dropped
72435	43325	29110

Engineered Feature Statistics

The MEANS Procedure

Variable	Label	Mean	Std Dev	Minimum	Maximum
car_age	Car Age (years as of 2020)	2.9217773	2.1032897	0	24.0000000
price_per_mile	Price per Mile Driven (£)	51.1027191	427.6872513	0.0055785	34500.00
efficiency_score	Fuel Efficiency Score (mpg per litre)	38.6031046	19.8273898	0.3000000	784.6666667

Encoded Variable Distributions

The FREQ Procedure

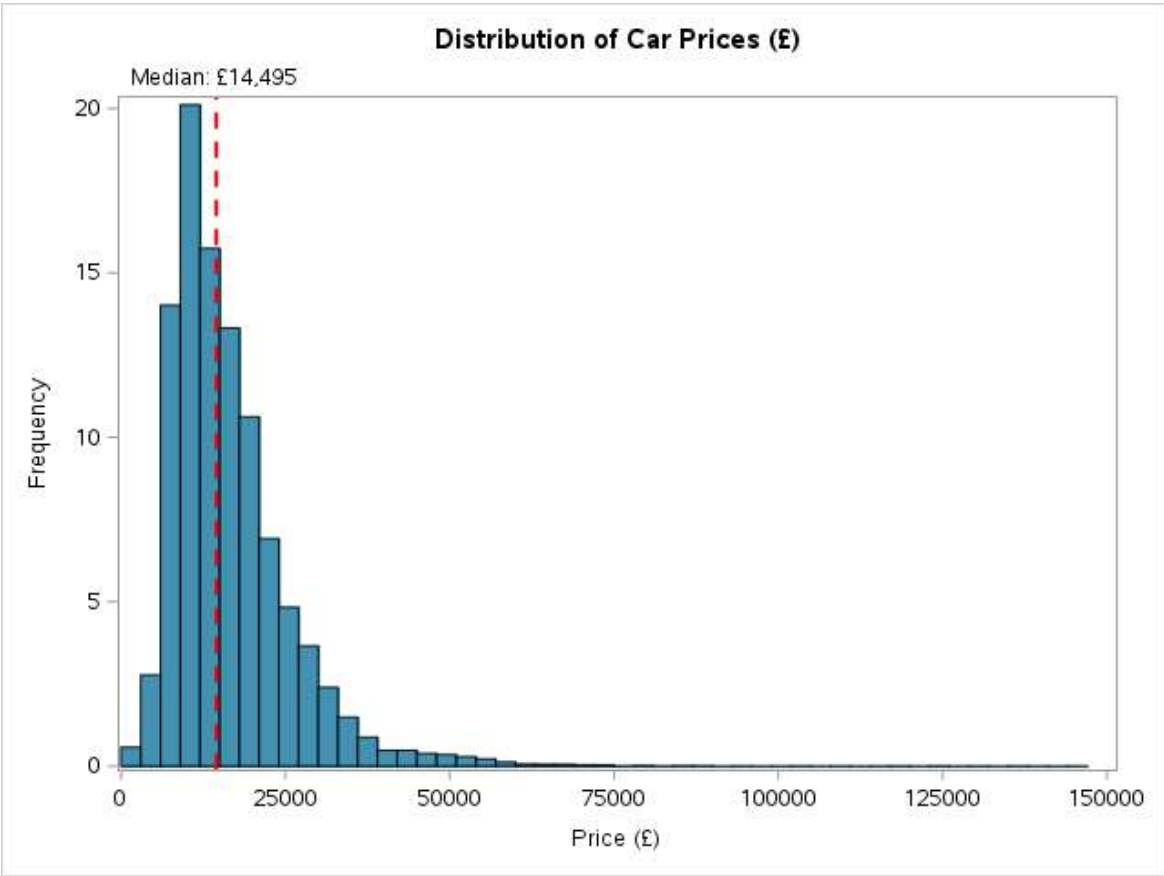
High Price Flag (1=Above Median, 0=Below Median)				
high_price	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	21694	50.07	21694	50.07
1	21631	49.93	43325	100.00

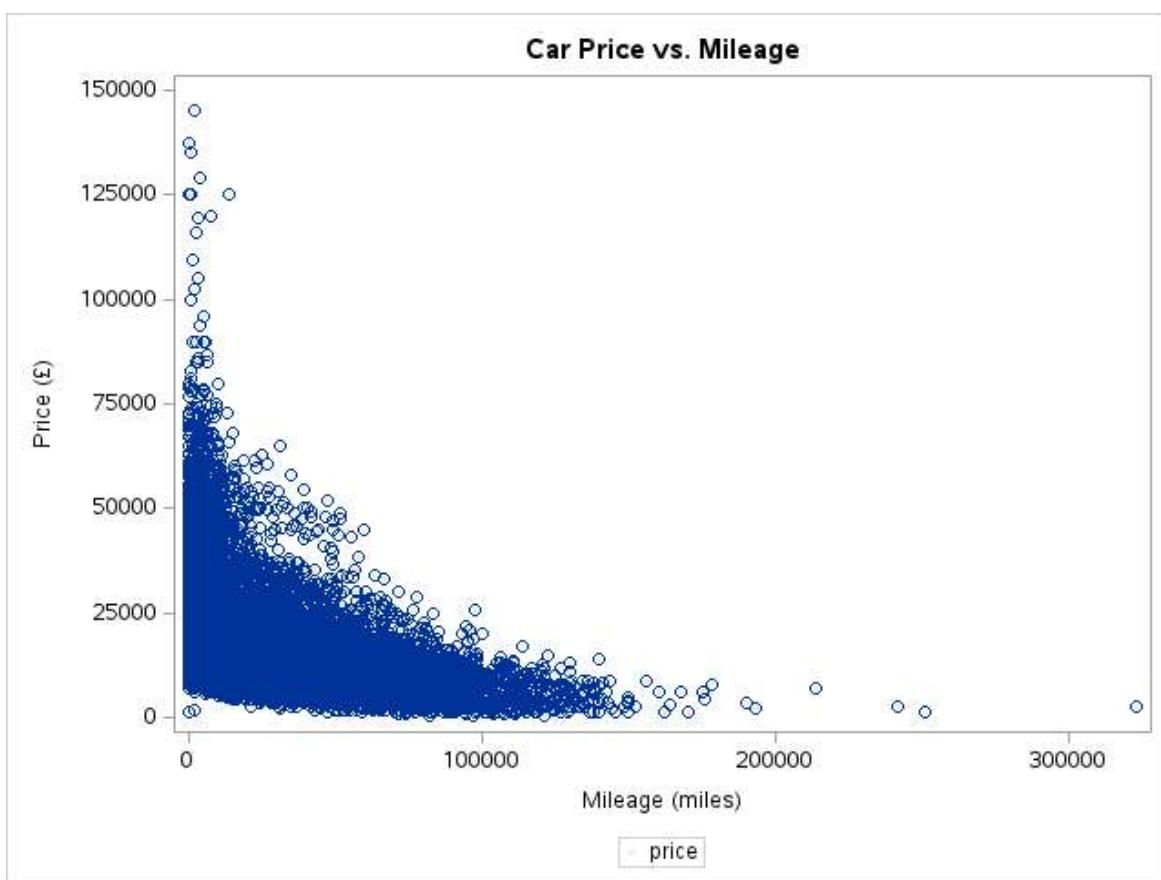
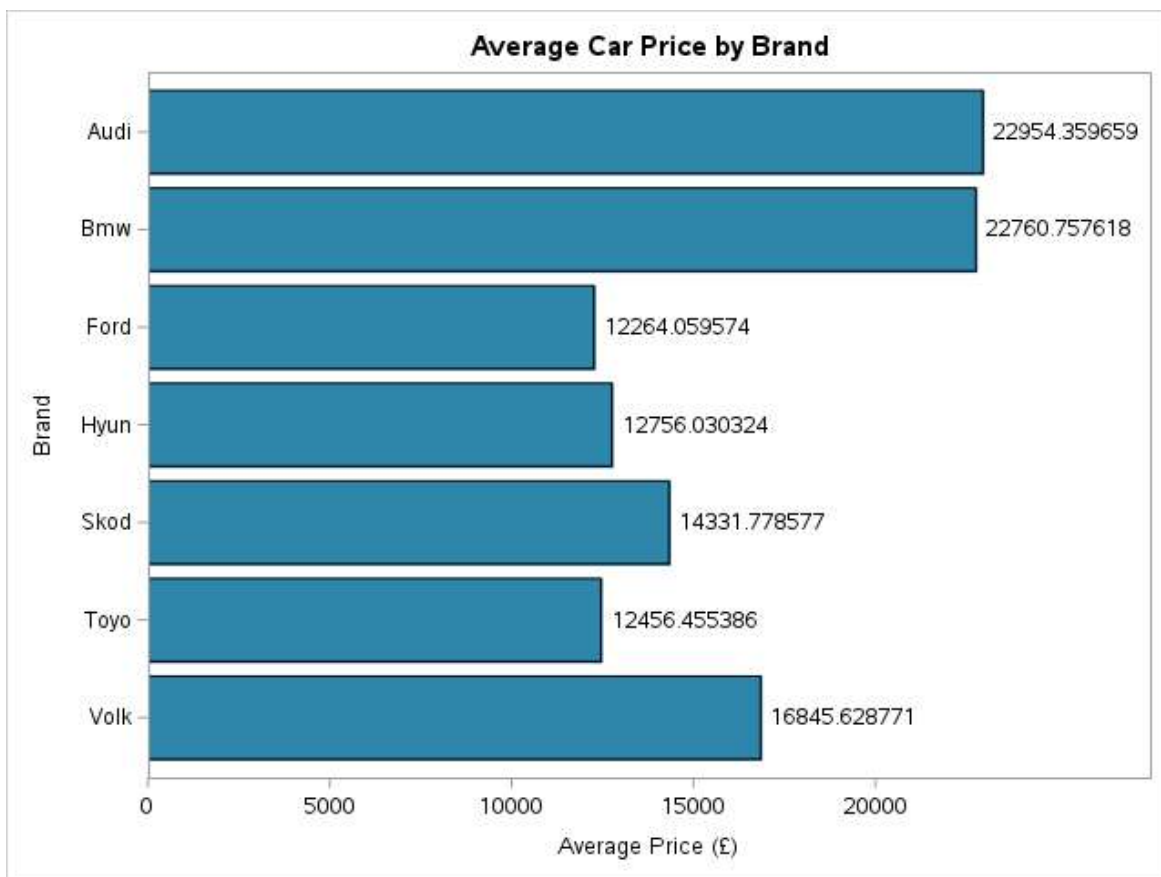
Make_num	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	19672	45.41	19672	45.41
1	6445	14.88	26117	60.28
2	6465	14.92	32582	75.20
3	10743	24.80	43325	100.00

trans_num	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	4	0.01	4	0.01
1	25743	59.42	25747	59.43
2	8415	19.42	34162	78.85

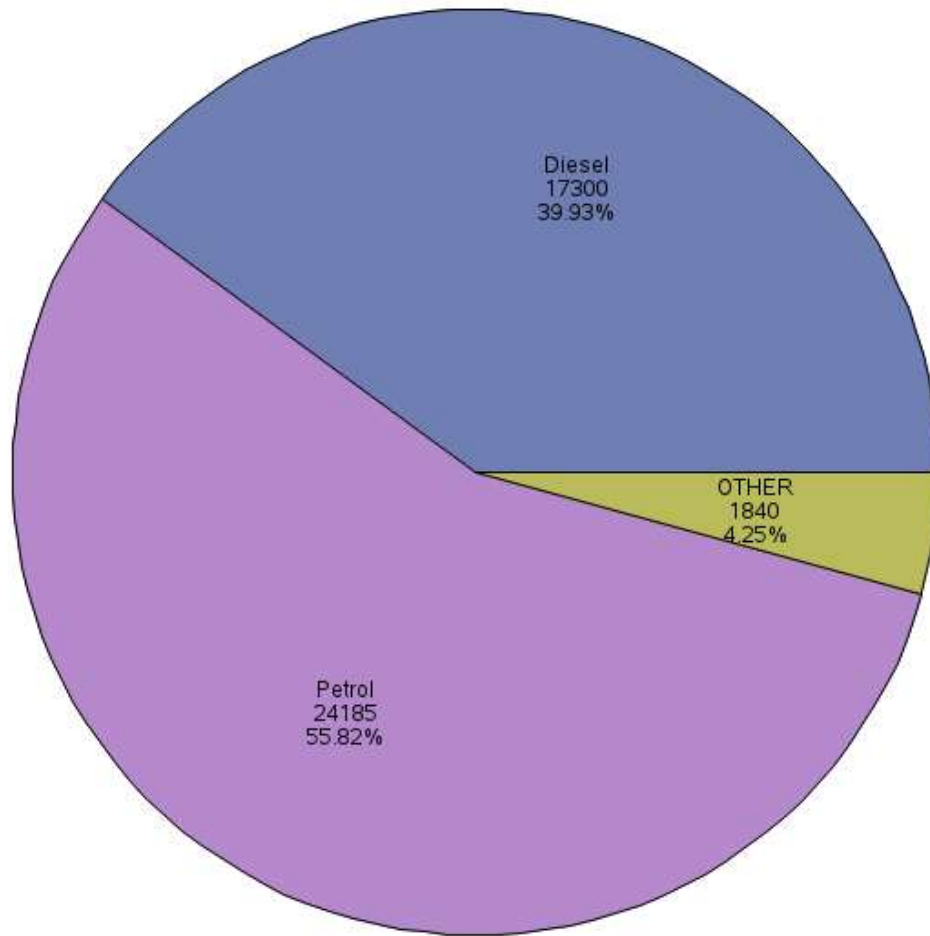
trans_num	Frequency	Percent	Cumulative Frequency	Cumulative Percent
3	9163	21.15	43325	100.00

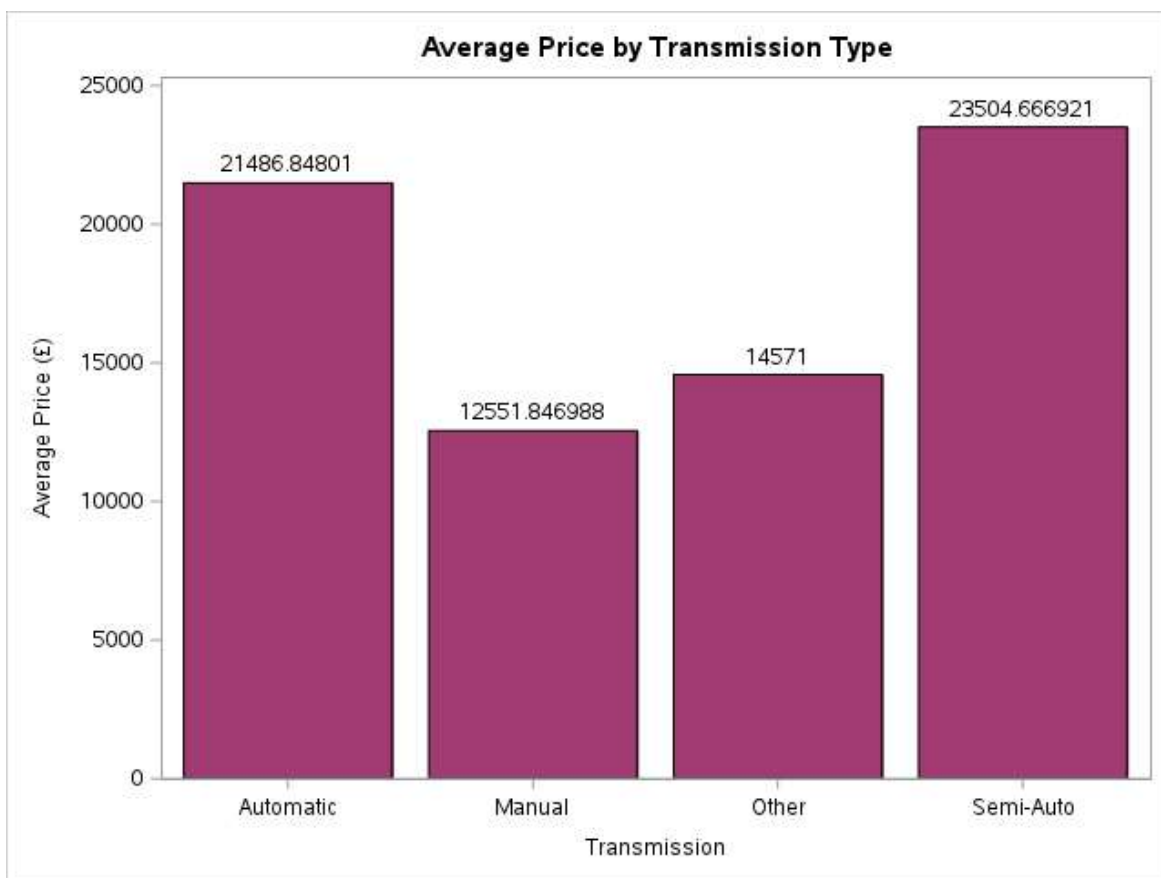
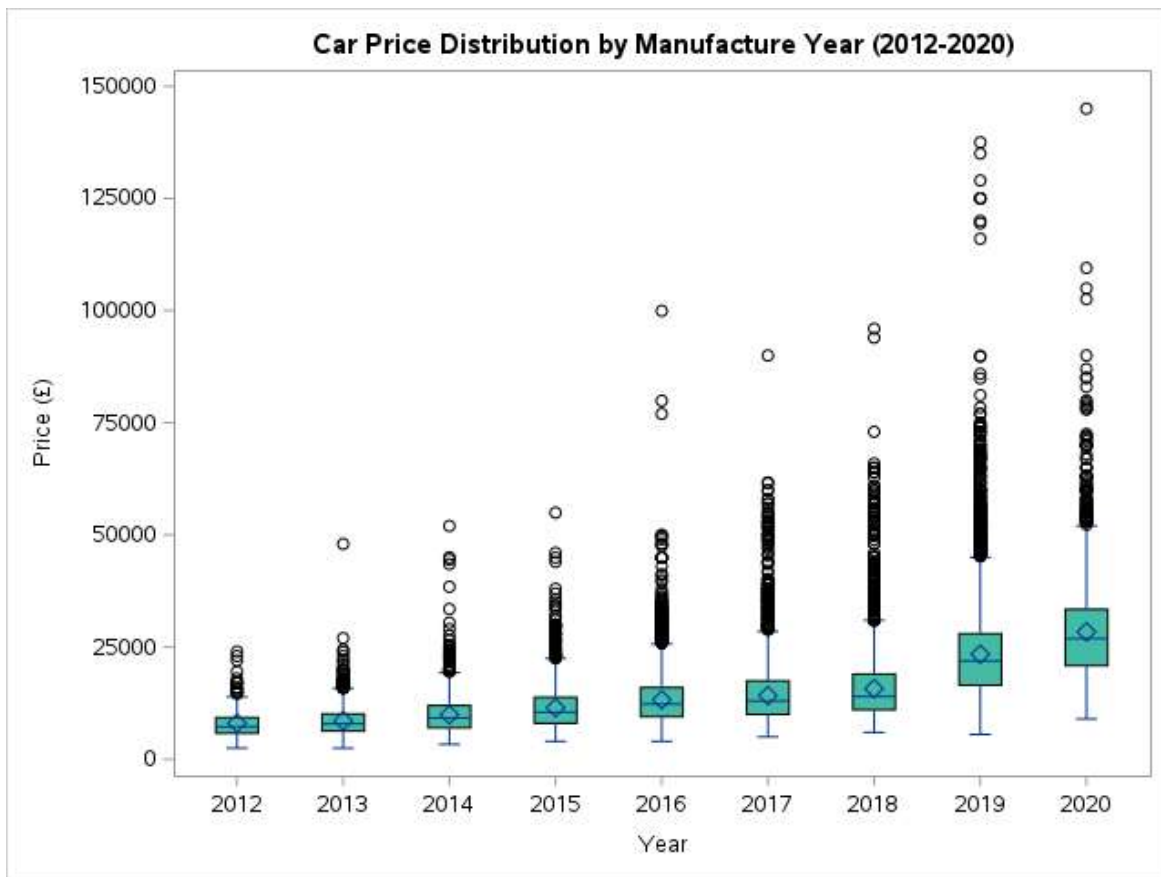
fuel_num	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	131	0.30	131	0.30
1	24185	55.82	24316	56.12
2	17300	39.93	41616	96.06
3	1709	3.94	43325	100.00

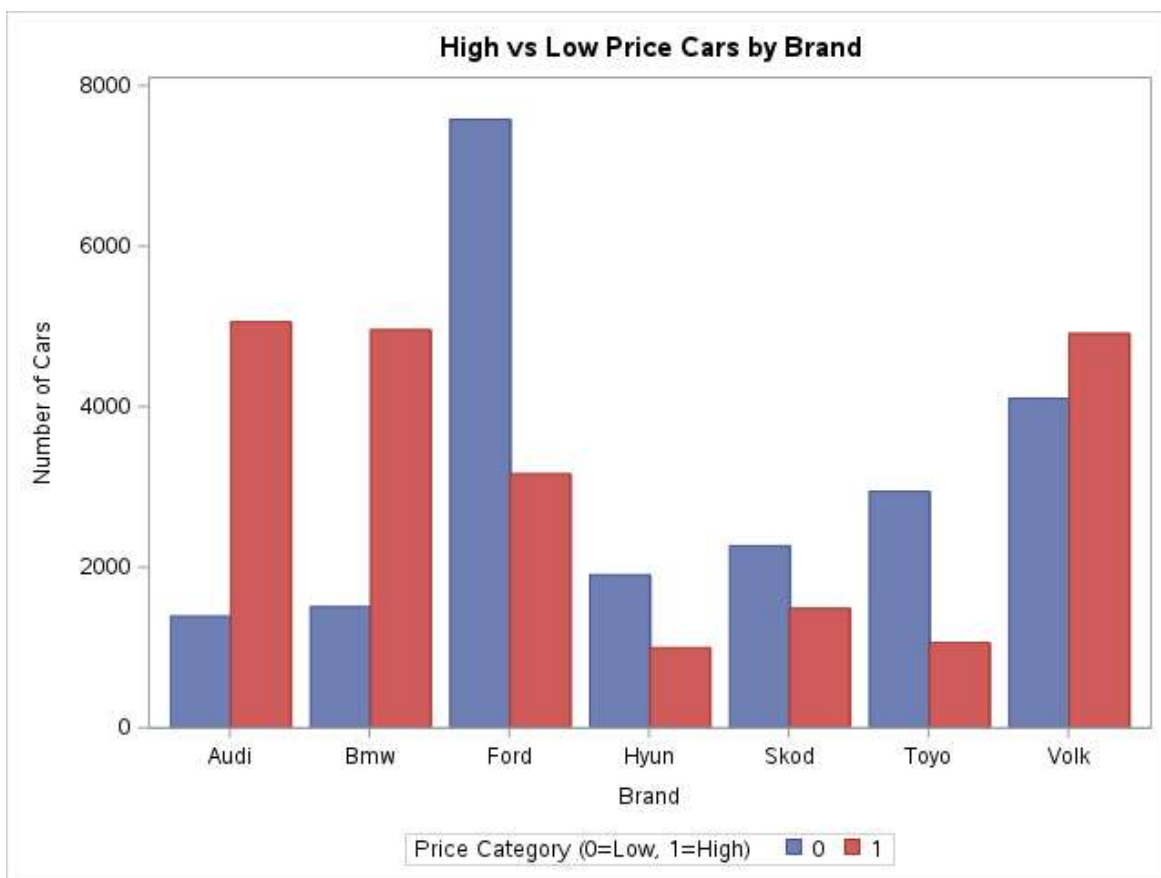
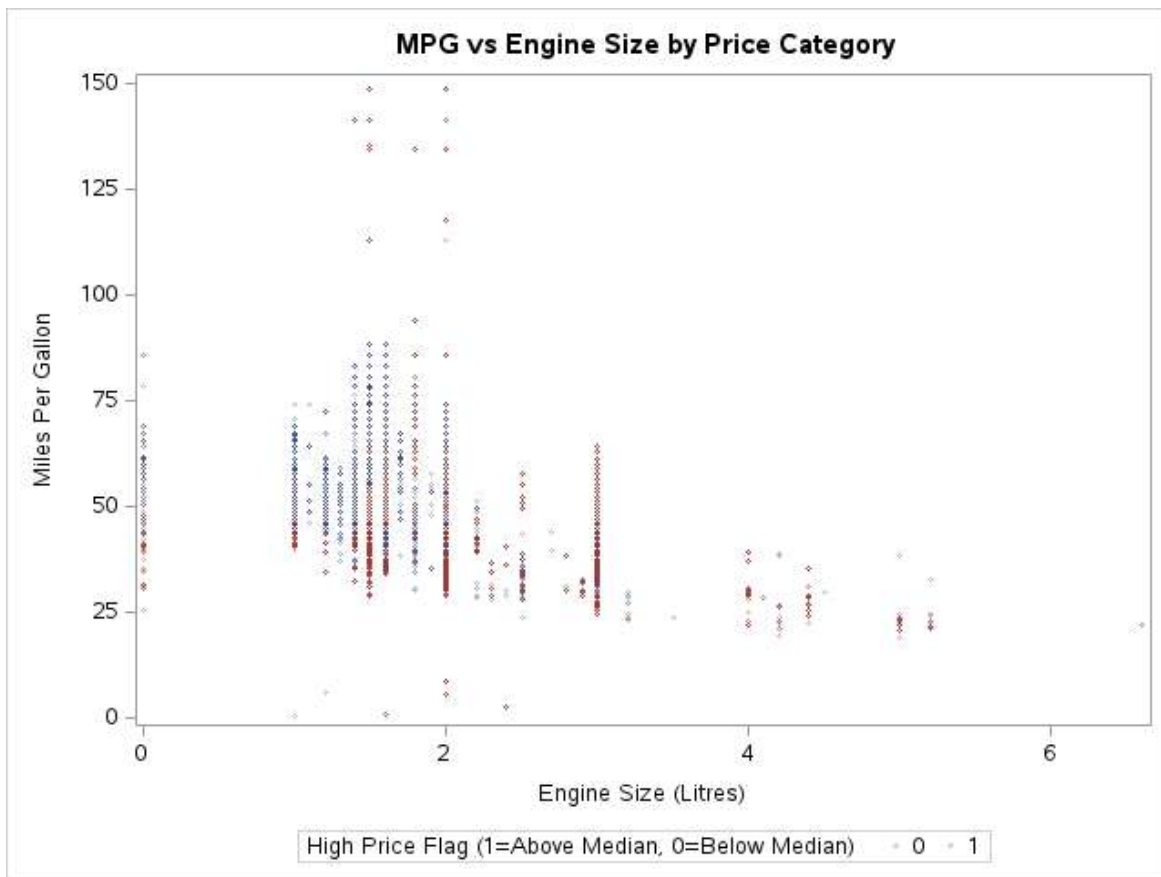




Fuel Type Distribution
SUM of Frequency Count by fuelType







High vs Low Price Cars by Brand

dataset	n
Test Set	8691
Training Set	34634

Logistic Regression - Training Results

The LOGISTIC Procedure

Model Information		
Data Set	WORK.CARS_TRAIN	
Response Variable	high_price	High Price Flag (1=Above Median, 0=Below Median)
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	34634
Number of Observations Used	34634

Response Profile		
Ordered Value	high_price	Total Frequency
1	0	17416
2	1	17218

Probability modeled is high_price=1.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

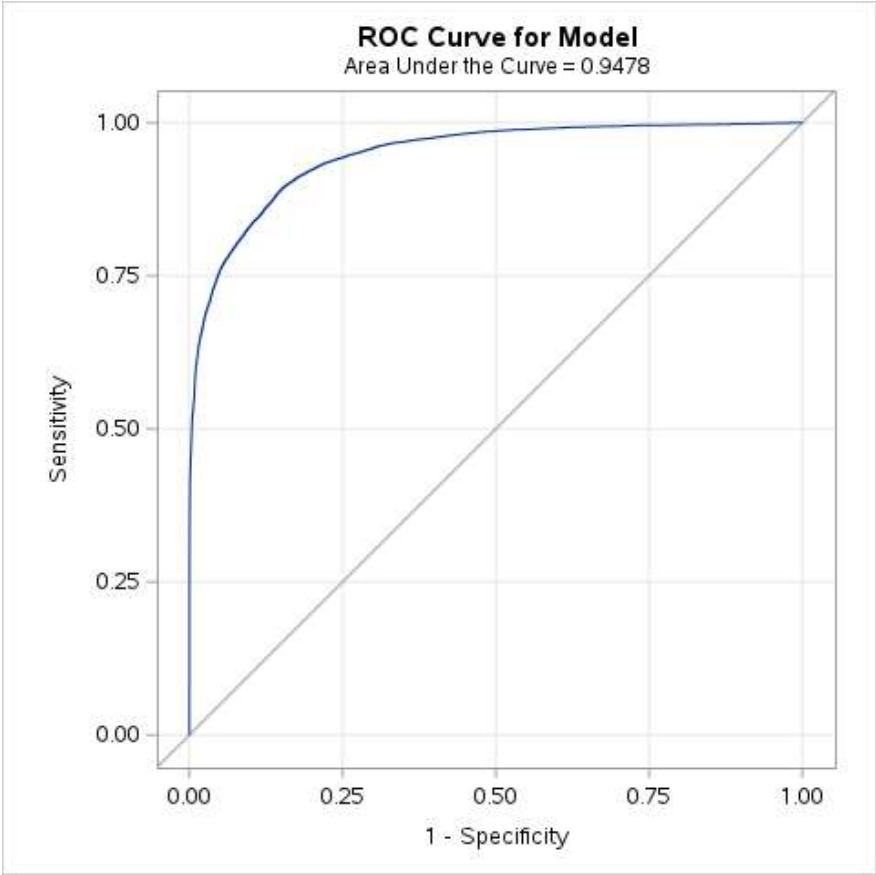
Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	48013.787	20346.728
SC	48022.240	20439.706
-2 Log L	48011.787	20324.728

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	27687.0590	10	<.0001
Score	18360.8210	10	<.0001
Wald	7795.9691	10	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-6.4840	0.1607	1627.3061	<.0001
car_age	1	-0.9469	0.0191	2448.8198	<.0001
mileage	1	-0.00004	1.567E-6	789.9762	<.0001
mpg	1	-0.00622	0.00336	3.4257	0.0642
engineSize	1	5.0406	0.1057	2274.8443	<.0001
tax	1	0.00553	0.000399	192.0255	<.0001
price_per_mile	1	0.000228	0.000090	6.3685	0.0116
efficiency_score	1	0.0121	0.00304	15.9113	<.0001
Make_num	1	0.1022	0.0138	54.9539	<.0001
fuel_num	1	0.0454	0.0531	0.7304	0.3928
trans_num	1	0.8256	0.0255	1050.9343	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
car_age	0.388	0.374	0.403
mileage	1.000	1.000	1.000
mpg	0.994	0.987	1.000
engineSize	154.563	125.646	190.136
tax	1.006	1.005	1.006
price_per_mile	1.000	1.000	1.000
efficiency_score	1.012	1.006	1.018
Make_num	1.108	1.078	1.138
fuel_num	1.046	0.943	1.161
trans_num	2.283	2.172	2.400

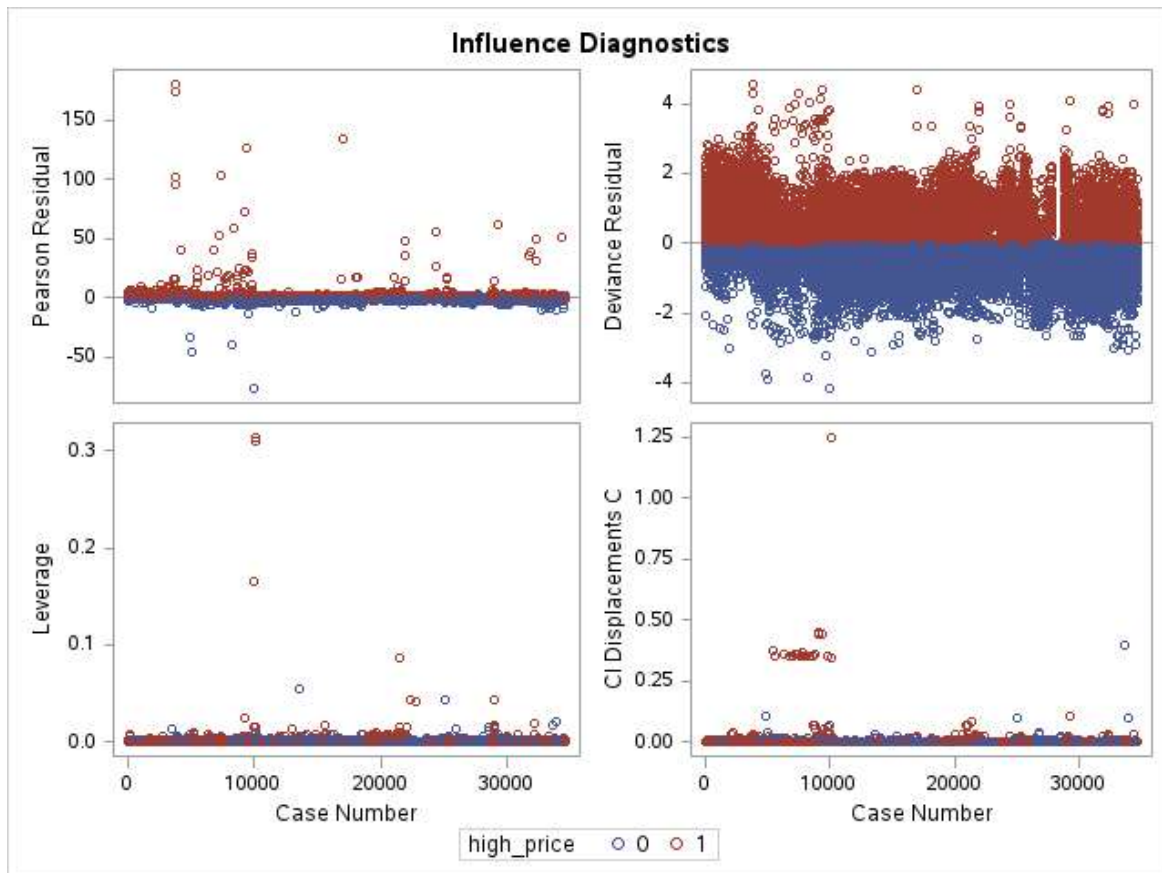
Association of Predicted Probabilities and Observed Responses			
Percent Concordant	94.8	Somers' D	0.896
Percent Discordant	5.2	Gamma	0.896
Percent Tied	0.0	Tau-a	0.448
Pairs	299868688	c	0.948

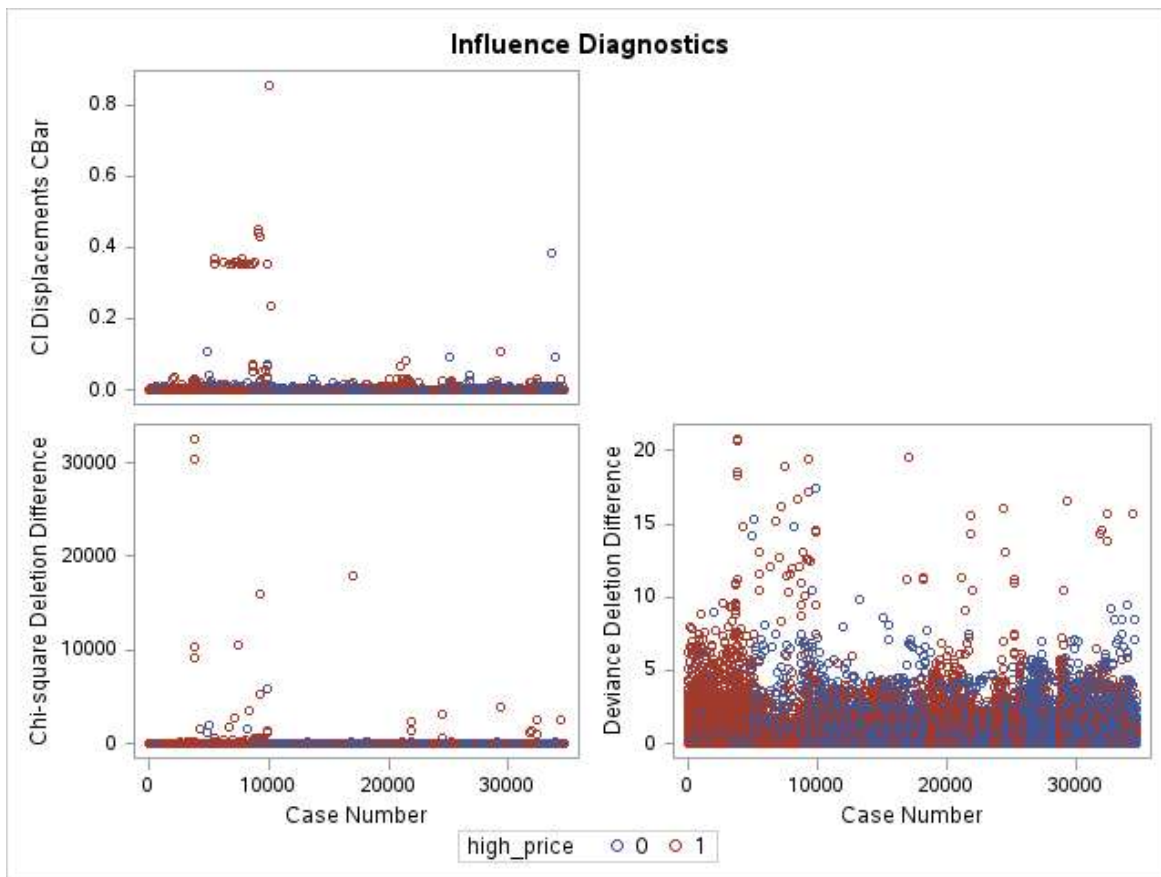


Partition for the Hosmer and Lemeshow Test					
Group	Total	high_price = 1		high_price = 0	
		Observed	Expected	Observed	Expected
1	3463	60	11.67	3403	3451.33
2	3463	79	90.27	3384	3372.73
3	3463	218	278.12	3245	3184.88
4	3463	568	633.16	2895	2829.84
5	3463	1300	1216.93	2163	2246.07
6	3463	1989	1991.00	1474	1472.00
7	3463	2788	2839.86	675	623.14

Partition for the Hosmer and Lemeshow Test					
Group	Total	high_price = 1		high_price = 0	
		Observed	Expected	Observed	Expected
8	3463	3302	3261.36	161	201.64
9	3464	3452	3433.61	12	30.39
10	3466	3462	3462.01	4	3.99

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
258.6242	8	<.0001





Logistic Regression - Test Set Scoring

The LOGISTIC Procedure

Fit Statistics for SCORE Data											
Data Set	Total Frequency	Log Likelihood	Error Rate	AIC	AICC	BIC	SC	R-Square	Max-Rescaled R-Square	AUC	Brier Score
WORK.CARS_TEST	8691	-2529.3	0.1331	5080.62	5080.65	5158.39	5158.39	0.552465	0.736679	0.9491	0.090087

Confusion Matrix - Test Set

The FREQ Procedure

Frequency	Table of high_price by pred_class		
	pred_class		
high_price(High Price Flag (1=Above Median, 0=Below Median))	0	1	Total
0	3802	476	4278
1	681	3732	4413
Total	4483	4208	8691

Confusion Matrix - Test Set

accuracy
86.69

New Car Price Category Predictions

Obs	year	mileage	engineSize	Make	transmission	P_1	Price_Category
1	2019	12000	1	Ford	Manual	0.33826	Low Price
2	2018	25000	2	Bmw	Automati	.	Low Price
3	2020	5000	3	Audi	Semi-Aut	.	Low Price